RC Design
Project 01

We can delete this since it no need

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1 How to Design

- Building Load contains : **gravity** and **lateral** Loads.

1.1 Convert Area Loads to Line Load on Beams

Slabs distribute their load to beam depending on slab span direction and support condition (EN 1992-1-1:2023, Cl. 5.3.2 [1]; IStructE Manual, Cl. 3.2).

The Tributary Width Method [IStructE Manual, Cl. 3.2]:

For a one-way slab, load goes to the beams on the short sides:

2 Introduction

This is Our Team Intro.

3 Detailed Design Calculations

3.1 Loading

3.1.1 Live Load

For imposed laods on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]. Is summary in Table 3.1.

Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	**Sub-category / Use**	q_k **[kN/m²]**	Q_k **[kN]**
Category A	Areas for domestic and residential activities		
	- Floors	1,5 to 2,0	2,0 to 3,0
	- Stairs	2,0 to 4,0	2,0 to 4,0
	- Balconies	2,5 to 4,0	2,0 to 3,0
Category B	Office areas	2,0 to 3,0	1,5 to 4,5
Category C	Areas where people may congregate		
C1	Areas with tables (e.g. schools, cafes, restaurants, reading rooms)	2,0 to 3,0	3,0 to 4,0
C2	Areas with fixed seats (e.g. theatres, cinemas, churches)	3,0 to 4,0	2,5 to 7,0 (4,0)
C3	Areas without obstacles for moving people (e.g. museums, entrance halls)	3,0 to 5,0	4,0 to 7,0
C4	Areas with possible physical activities (e.g. dance halls, gyms)	3,5 to 5,0	3,5 to 7,0
C5	Areas susceptible to large crowds (e.g. concert halls, sports halls with stands)	5,0 to 7,5	3,5 to 4,5
Category D	Shopping areas		

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Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	**Sub-category / Use**	q_k [kN/m²]	Q_k [kN]
D1	Areas in general retail shops	4,0 to 5,0	3,5 to 7,0 (4,0)
D2	Areas in department stores	4,0 to 5,0	3,5 to 7,0
Category E	Storage and industrial activities (general storage, accessible areas)	7,5	7,0

And this is the imposed loads on second floor that we assumed. which shown in Table 3.2.

Table 3.2: Imposed Loads On Second Floor

No.	**Room / Area**	**Category (EN 1991-1-1)**	q_k **(kN/m2)**
1	Receptionist/ Reception	B	3
2	Workstation / Open office	B	3
3	Meeting Room	C1	3
4	President Office	B	3
5	Vice President Offices	B	3
6	Accounting	B	3
7	General Sale & Administration	B	3
8	Copy / Printer Area	B	3
9	Storages	E1	5
10	Lift Lobby / Lift Area	C3	4
11	Stair (Main Stair)	A	4
12	Restroom (Men)	B	3
13	Restroom (Women)	B	3
14	—	—	—
15	—	—	—
16	—	—	—

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Table 3.2: Imposed Loads On Second Floor

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m2)
17	Entrance	C3	4
18	Emergency Stair	A	4
19	Server& Technical Room	E1 (conservative)	5
20	Kitchen / Pantry	C1	3
21	Library	C3	4
22	Share Space	C3	4

3.1.2 Dead Load

According to EN 1991-1-1:2025, Table A.1 [2]; EN 1991-1-1:2002, Cl. 4 [3]:

Table 3.3: Dead Load Typical Value

Source	Typical Value
Reinforced concrete slab (200mm thick)	$0.2 \times 25 = 5.0\ \mathrm{kN/m^2}$
Screed/floor finish	$1.5\ \mathrm{kN/m^2}$
Ceiling + services (MEP)	$0.5\ \mathrm{kN/m^2}$
Partitions (treated as permanent)	$1,0\ \mathrm{kN/m^2}$
Total G_k	$8.0\ \mathrm{kN/m^2}$

References

[1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.

[2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.

[3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.

[4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.

[5] CEN, *Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, EN 1991-1-4:2005+A1, English version, April 2010.

Note that : We may use the referennces provide by professor (check in telegram group) if it still not enought we can add more maybe.

Appendices